
Residual Calibration for Property Guidance: A Two-Modality Experimental Study

Independent Research Study

Abstract

Nonlinear properties of denoised means can misestimate quantities needed for guidance. We test a 28-parameter residual correction on frozen CIFAR-10 and continuous WikiText-2 flow models. It avoids property Hessians but still differentiates through the generator. We compare property heads, uncertainty, and postprocessing using strength sweeps and three independent sampling seeds. Residual calibration reduces target error versus a tuned plug-in by 82.5% on clipped images and 50.6% on decoded text representations. However, a direct head wins on text, image postprocessing is competitive at half the latency, and fluent controlled text is not demonstrated. The contribution is a reproducible empirical extension, not a new general guidance framework.

1 Introduction

Guidance evaluates a desired property on a denoised estimate, even though a nonlinear property of a conditional mean need not equal its conditional expectation. We investigate whether a small supervised correction improves practical control while leaving a pretrained generative model fixed.

The intervention learns the residual between a known clean property $f(X)$ and its value on the denoised estimate $f(m)$. A 28-coefficient head combines compact state statistics with a clean-endpoint factor. We test the same construction on image variance and adjacent-vector distance in continuous text, comparing time-only calibration, direct prediction, uncertainty, and elementary postprocessing. The experiment isolates control, output conversion, general quality, and computation. Originality priority for the exact combination is not established.

2 Related Work and Scope

DPS uses denoised plug-in measurement guidance [Chung et al., 2023]; LGD analyzes point-estimate error and uses Monte Carlo approximations [Song et al., 2023]. TMPD derives posterior moments and tractable likelihood approximations [Boys et al., 2024]. Readout Guidance trains lightweight property heads on frozen diffusion features [Luo et al., 2024]; DEFT learns a small conditional correction and can avoid generator backpropagation [Denker et al., 2024]. Moment correction, small heads, and frozen backbones are established.

Our scalar direct head is readout-style, not a reproduction of the published spatial architecture. Our plug-in transfers the DPS principle to a Gaussian-path flow, not its original sampler. The residual can compensate for both nonlinear-property bias and denoiser error. Its value must therefore be demonstrated against simple calibrated predictors, not inferred from a posterior identity or a lower regression loss.

3 Setup and Algorithm

3.1 Frozen Generators and Data

Let X denote clean data, $Z \sim \mathcal{N}(0, I)$ independent noise, $t \in (0, 1)$ the data fraction, and x the current sampler state. The straight flow-matching path and training target [Lipman et al., 2023] are

$$X_t = tX + (1 - t)Z, \quad v^* = X - Z, \quad m_\theta(x, t) = x + (1 - t)v_\theta(x, t). \quad (1)$$

Image class conditioning is fixed and suppressed in the formulas. The ideal m is $\mathbb{E}[X \mid X_t = x]$; the learned m_θ only approximates it. Images use a class-conditioned 2.63M-parameter U-Net, trained for 20k updates on a fixed 45k/5k CIFAR-10 [Krizhevsky, 2009] training/validation split, with the official 10k test set. Text uses an unconditional 3.316M-parameter Transformer trained for 30k updates. Its state contains 32 positions of 32-dimensional fixed RMS-normalized code vectors from a training-only 4,096-token BPE vocabulary. WikiText-2 [Merity et al., 2016] windows number 104,703/10,984/12,512 across splits. We use pretrained generators with model-training seed 0 and freeze them throughout calibration and sampling. Their weights and configurations are supplied as local model assets; no new generator training is claimed.

3.2 Property Likelihood and Residual Calibration

For scalar target y , define $L_y(X) \propto \exp[-(f(X) - y)^2 / (2s^2)]$. Exact conditional guidance requires $h_t(x) = \mathbb{E}[L_y(X) \mid X_t = x]$, generally unavailable. We approximate it by a Gaussian likelihood:

$$\log \hat{h}_t(x) = -\frac{1}{2} \left\{ \frac{(y - \hat{\mu}_t(x))^2}{s^2 + V(t)} + \log[s^2 + V(t)] \right\}, \quad \hat{\mu}_t = f(m_\theta) + C_\phi(x, m_\theta, t). \quad (2)$$

On 768 training examples at six noise levels, regress $f(X) - f(m_\theta)$ with

$$C_\phi = (1 - t)\phi(x, m_\theta, t)^\top \beta. \quad (3)$$

The 28 coefficients include an intercept. Features are time, RMS values, means, one state/velocity cosine, $f(m)$, and time interactions (Appendix A). A training-only holdout selects ridge regularization; 384 separate validation examples assess prediction. The factor $1 - t$ forces the residual to vanish at the clean endpoint. Optional $V(t)$ is interpolated out-of-fold training residual MSE, *not* state-specific posterior variance. Mean-only calibration sets $V = 0$.

3.3 Actual Sampling Rule

For the ideal Gaussian bridge, velocity and score satisfy $v_t = x/t + (1 - t)\nabla \log p_t(x)/t$. Thus the guidance increment is

$$d = \lambda \frac{1 - t}{t} \nabla_x \log \hat{h}_t(x), \quad \tilde{v} = v_\theta + \min\left(1, \frac{\text{RMS}(v_\theta)}{\max(\text{RMS}(d), \epsilon)}\right) d. \quad (4)$$

We activate it on $t \in [.2, .95]$ and integrate with 16 uniform Heun intervals on $[.001, .999]$. Autograd includes the full derivative through m_θ and C_ϕ . The cap uses $\epsilon = 10^{-12}$. Each guided sample uses 32 generator forwards and 24 input backwards; head reductions add no generator calls. Approximation, strength, gating, and clipping prevent an exact conditional-sampling claim. Two Heun field evaluations are recomputed per interval; the corrector state is not reused as the next predictor state.

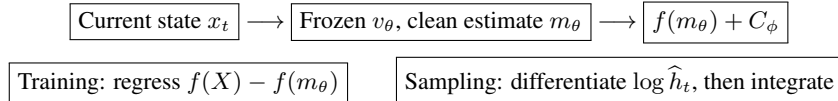


Figure 1: Only the small calibration head is fitted. Sampling differentiates through both the head and frozen generator. Output conversion is evaluated separately.

4 Experimental Design

4.1 Control Tasks, Baselines, and Measurements

Images use centered pixel variance $f(x) = d^{-1} \sum_j (x_j - \bar{x})^2$; text uses adjacent-code-vector distance $f(x) = ((L-1)D)^{-1} \sum_{i,j} (x_{i+1,j} - x_{i,j})^2$. Targets alternate between the training-subset 25th and 75th percentiles; $s = .25 \text{SD}_{\text{train}}(f)$. Each image target group has the same class histogram. Text’s property includes all 32 positions, including special tokens and positions after EOS. It is a representation statistic, not a semantic or fluency objective.

We compare unguided sampling; the plug-in $C = V = 0$; six-knot time-only $C(t)$; residual mean calibration; residual mean plus $V(t)$; and a direct 28-coefficient predictor of $f(X)$ with identical features and training examples. A post-hoc positive rescaling control exactly matches the raw quadratic property with no backwards. Clipping can spoil exact image control; positive scaling approximately preserves nearest-code-vector identities for equal-norm code vectors. This elementary baseline tests whether the chosen task requires guidance at all.

Report image target MAE after clipping, full unbiased Inception KID, precision/recall, and an independent frozen classifier’s class accuracy (89.22% validation accuracy). Text reports raw and nearest-code-vector property error, frozen GPT-2 perplexity, length, distinct-2, malformed Unicode, and visible-prefix diagnostics. The nearest-code-vector statistic still includes all positions; the human-readable decoder removes special tokens and stops at EOS. No metric alone establishes fluency.

4.2 Training and Diagnostic Validation

Each modality supplies 768 training examples at six noise levels, giving 4,608 corruption pairs. A training-example-disjoint holdout selects ridge regularization, and five-fold training predictions calibrate optional error variance. Another 384 examples at the same levels give 2,304 validation pairs. These diagnostics are separate from generated-sample selection. No target value, clean state, or sampled noise is an inference feature.

Table 1: Property-prediction diagnostics on independent corruption validation. Lower MSE/NLL is better. NLL uses a time-dependent error scale; it is not an exact posterior-likelihood evaluation.

Predictor	Image MSE	Image NLL	Text MSE	Text NLL
Plug-in	.008344	−1.868	1.1924	9.970
Time mean	.002154	−3.014	.03278	−1.701
Residual mean	.001988	−2.849	.01504	−2.077
Direct predictor	.002016	−2.645	.01529	−2.026

The state residual improves property MSE relative to time-only calibration but worsens image Gaussian NLL. Thus better scalar prediction need not yield better uncertainty or a useful sampling gradient. The direct control changes both the regression target and endpoint parameterization; it is not an isolated ablation of the factor $1 - t$.

4.3 Controls, Selection, and Replication

The seven methods test complementary explanations: plug-in tests the need for calibration; time-only correction tests sample dependence; the direct head tests residual parameterization; added variance tests uncertainty weighting; and postprocessing tests whether guidance is needed for the chosen property. All derivative-guided variants share the integration grid, interval, cap, and search budget.

Development uses 160 generated samples per modality, shared noise seed 29001, and six strengths. Each method selects converted-output MAE; quality and diversity are reported separately. The frozen comparison then uses seeds 39001–39003, 400 samples per method/seed. Image references are a fixed 400-image official-test subset, independent of generator samples but not matched to the property targets. KID therefore measures marginal quality. Small-sample uncertainty and one fixed generator per modality limit generalization; none of these results measures robustness to retraining.

5 Generation Results

Frozen selection. Each method chooses the lowest delivered-representation MAE among $\lambda \in \{.125, .25, .5, 1, 2, 4\}$ on 160 development samples. There is no hidden quality constraint or claim of global optimality; several choices lie at the grid boundary. All image guided methods select 4. Text selects .125 (plug-in), 2 (time/state mean), and 4 (state variance/direct head). Choices are frozen before replication; tables give sampling-seed uncertainty.

Method	Output MAE	Raw MAE	KID $\times 100$	Class acc.	ms/sample
Base	0.0862 ± 0.0029	0.0867 ± 0.0029	5.85 ± 0.23	0.547 ± 0.029	11.7 ± 0.1
Post-hoc	0.0055 ± 0.0006	0.0000 ± 0.0000	5.76 ± 0.27	0.531 ± 0.029	11.7 ± 0.1
Plug-in	0.0280 ± 0.0002	0.0291 ± 0.0002	5.76 ± 0.21	0.525 ± 0.016	23.0 ± 0.5
Time mean	0.0096 ± 0.0001	0.0094 ± 0.0001	5.71 ± 0.22	0.529 ± 0.016	24.0 ± 0.7
State residual mean	0.0049 ± 0.0001	0.0042 ± 0.0001	5.72 ± 0.22	0.527 ± 0.019	24.7 ± 0.7
State mean + var.	0.0077 ± 0.0003	0.0070 ± 0.0003	5.87 ± 0.20	0.520 ± 0.016	24.9 ± 0.5
Direct state mean	0.0052 ± 0.0001	0.0045 ± 0.0001	5.75 ± 0.21	0.522 ± 0.022	24.8 ± 0.9

Table 2: Images: clipped image variance. Mean $\pm$ sample SD over three sampling-noise seeds, 400 samples each; one frozen training seed.

Method	Output MAE	Raw MAE	GPT-2 PPL	Distinct-2	ms/sample
Base	0.1254 ± 0.0032	0.2279 ± 0.0025	9329 ± 369	0.941 ± 0.005	3.3 ± 0.1
Post-hoc	0.1254 ± 0.0032	0.0000 ± 0.0000	9329 ± 369	0.941 ± 0.005	3.6 ± 0.3
Plug-in	0.1236 ± 0.0030	0.1174 ± 0.0032	5236 ± 262	0.865 ± 0.005	8.8 ± 0.7
Time mean	0.1039 ± 0.0036	0.1410 ± 0.0013	8251 ± 117	0.920 ± 0.002	9.5 ± 0.6
State residual mean	0.0610 ± 0.0022	0.1588 ± 0.0027	9270 ± 222	0.940 ± 0.004	10.6 ± 1.0
State mean + var.	0.0605 ± 0.0017	0.1474 ± 0.0027	9259 ± 248	0.940 ± 0.004	10.3 ± 0.7
Direct state mean	0.0534 ± 0.0021	0.1793 ± 0.0030	9259 ± 49	0.940 ± 0.003	10.4 ± 0.5

Table 3: Continuous text: decoded 32-position embedding adjacency. Mean $\pm$ sample SD over three sampling-noise seeds, 400 samples each; one frozen training seed.

Control improves; residual structure does not win uniformly. Residual mean calibration reduces clipped-image MAE by 82.5% versus the tuned plug-in and 48.6% versus time-only calibration. Its advantage over the direct predictor is small (.0049 versus .0052). Text’s decoded-representation MAE falls 50.6% versus plug-in, despite worse raw error; the direct predictor is better (.0534 versus .0610). Added variance worsens image control and provides little text benefit. These are control results, not evidence of improved image fidelity or fluent text.

Validation control-quality tradeoff: 160 samples, shared noise seed

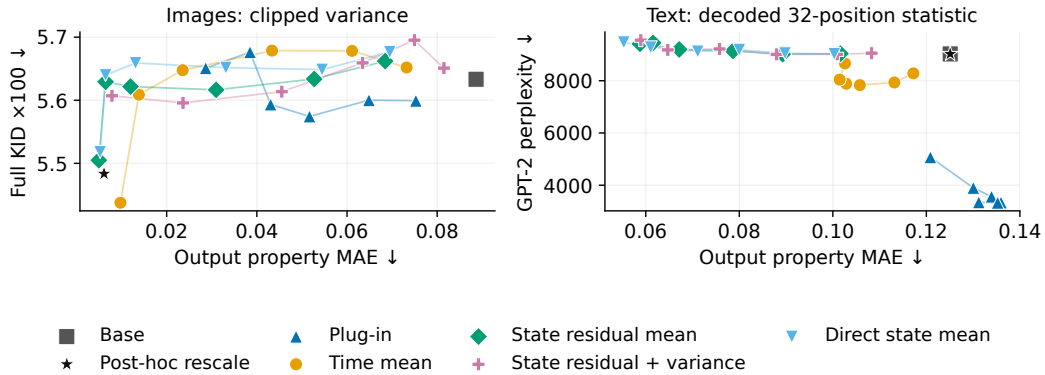


Figure 2: Development trade-offs across the common strength grid. These curves selected final settings; they are not independent replication evidence.

6 Discussion and Limitations

The useful result and the limiting baselines. The calibration intervention improves target control over a tuned plug-in, but does not justify a general residual-architecture advantage: the direct head wins on text. Image postprocessing reaches MAE .0055 at 11.7 ms/sample, versus residual calibration’s .0049 at 24.7 ms. The latter’s KID change is small relative to sampling variation, and class accuracy is .527 versus the base’s .547. Text postprocessing leaves all 1,200 decoded token sequences identical to the base despite near-zero raw-property error. Plug-in guidance lowers PPL from 9329 to 5236 while reducing distinct-2 from .941 to .865; fixed-index outputs remain incoherent. No method demonstrates a universal control–quality–cost improvement. The parameter budget is small, but inference is not free. Calibration preparation uses frozen-network evaluations of corrupted training data; the property-only extraction workload is 4,608 training and 2,304 diagnostic model-example evaluations per modality. These counts specify workload, not a newly measured extraction runtime. Ridge fitting takes approximately 10 ms on cached features; that excludes cache construction. Guided generation still needs 24 full input-backpropagation operations per sample. We report these separately rather than assigning an arbitrary forward-equivalent NFE or claiming a speedup over TMPD or DEFT without implementing them.

The main scientific distinction is between estimating $\mathbb{E}[f(X) \mid x_t]$ and controlling the actual delivered representation. The learned head is trained on forward-corrupted real examples, while guidance visits altered trajectories. The direct head may extrapolate differently; the clean-endpoint factor does not guarantee on-trajectory calibration. Optional time-dependent error variance can suppress useful guidance and should not be called a calibrated state posterior. Small image KID and kNN metrics also have substantial sampling uncertainty.

This pilot has one generator training seed, modest generation counts, simple nonsemantic properties, and an already weak text model. Independent noise seeds quantify sampling variability conditional on those fixed generators; they do not quantify training-seed robustness. The property-strength sweeps and CPU pilots are exploratory and excluded from replication claims. Reference datasets were not treated as newly sealed holdouts; fresh noise measures sampling variability conditional on fixed generators.

The work supplies a runnable intervention, time-only and uncertainty ablations, a direct-predictor control, and measured failure cases across two continuous modalities. These controls are not full reproductions of every cited system. A broader benchmark should implement the closest property-guidance methods on the same task and evaluate more informative text properties. Larger generators, more training seeds, and semantic targets would test external validity; their outcomes are unknown.

Conclusion. A tiny residual head is a concrete way to test property-likelihood calibration without constructing a property Hessian. The experiments should be read as evidence about its measured trade-offs and limits, not as proof that learning guidance or adding uncertainty is new, that a Gaussian h approximation is exact, or that denoising improvements imply better generation. The measured control gains support calibrating the guidance statistic, while the direct-head and postprocessing controls limit claims about the residual architecture and its practical advantage.

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A Complete Features, Fitting, and Sampling

For each sample, flatten all nonbatch axes. The eleven raw features are $t, t^2, t^3, \text{RMS}(x), \text{RMS}(v), \cos(x, v), \text{mean}(m), \text{RMS}(m), \text{std}(m), \text{mean}(v), f(m)$. Append the last eight multiplied by t and by t^2 , giving 27 standardized inputs and an intercept. Means and scales are training-only. For residual fitting, multiply the full design by $1 - t$, then solve ridge normal equations against $f(X) - f(m)$. Candidate regularization values are $10^{-6}, 10^{-4}, .001, .01, .1$; a disjoint 20% training-example holdout selects the coefficient. The final head refits all 768 examples. Five training-example-disjoint folds estimate predictive squared error by time. Repeated corruptions of one clean example stay together in a split. Times are .05, .2, .4, .65, .85, .97. Time-only C and all V are interpolated with zero clean-endpoint values; the state residual uses its analytic $(1 - t)$ factor, while the direct predictor has no such constraint. Ridge strength is selected once before five-fold error estimation; the coefficient fits are out of fold, but hyperparameter selection is not nested, so the variance calibration can remain optimistic.

The time-only residual is the mean residual in each time bin, linearly interpolated. Its optional variance in the CPU pilot is centered in-sample residual variance, unlike the state’s out-of-fold squared-error estimate. The principal generation comparison uses mean-only variants; any uncertainty comparison retains this caveat.

1. Fit the head from frozen-model corruptions of training examples; fix targets, observation scale, and selected sampling settings before final evaluation.
2. Draw a Gaussian initial state and, for images, fix the class label. Initialize at $t = .001$.
3. At each Heun evaluation, predict v_θ and $m_\theta = x + (1 - t)v_\theta$.
4. In the active interval, construct Eq. (2), differentiate its sum over the independent batch, and apply Eq. (4). Outside it, use v_θ alone.
5. For step Δt , compute $a = \tilde{v}(x, t)$, $b = \tilde{v}(x + \Delta t a, t + \Delta t)$, then set $x \leftarrow x + \Delta t(a + b)/2$.
6. Clip image values to $[-1, 1]$ for delivery/evaluation. For text, find nearest code vectors, compute representation metrics, and decode the prefix before the first EOS after removing PAD/BOS/EOS tokens.

The ordinary Gaussian log-likelihood includes the variance log determinant. Current V is time-only, so its state derivative is exactly zero. We do not silently drop derivatives of a learned state-dependent variance. The target multiplier and RMS cap apply identically to all derivative-guided controls.

B Development Protocol, Full Tables, and Replication

Development and stopping. Generation validation uses 160 outputs per cell and shared noise seed 29001. Image class $(i//2) \bmod 10$ balances classes within the alternating property-target groups. The common six-strength sweep is development, not a preregistered discovery test. Two image validation jobs supply 18+14 cells; one text job supplies 32 cells. The final 42 cells comprise seven methods, two modalities, and three new sampling seeds, 400 samples each. No final result is used to retune a head or strength.

The protocol fixes the image test-reference subset with seed 8129, and uses independently generated seeds 39001–39003. References are not class/target-matched conditional distributions, so KID is an unconditional marginal-quality diagnostic. For text, a frozen local GPT-2 scores decoded sequences; its context resets per sequence, BOS is prepended, ending EOS is unscored, and empty samples are reported separately. Mean length, distinct-n, replacement characters, duplicates and repeats accompany PPL. All empty/unscorable cases remain in the raw records.

Full tables, including the 64-cell property development sweep, per-seed final metrics, and paired effects, are machine-readable in the accompanying analysis directory. The manuscript uses the full unbiased polynomial-MMD KID. KID subset SD is not a confidence interval. Bootstrap intervals resample paired generated examples within each noise seed and remain conditional on the fixed generator and chosen protocol.

Additional diagnostics. The independent paired-response experiment below uses the same noise for both property targets. Visible-prefix text diagnostics follow the actual decoder and avoid comparing the prefix statistic to a target defined on all 32 positions. Source/fit audits check held-out predictions against cached values within 1.02×10^{-6} and verify finite derivatives. Training and generation metrics are never substituted for one another.

C Counterfactual Target Responses

A separate diagnostic uses 32 independent noise/label pairs per modality and generates both target values from each identical initial state. Its noise seed is 9921, strength .25, and method list is fixed to plug-in, time mean, and state mean. This isolates target responsiveness from differences in initial noise. It remains a development diagnostic and measures a decoded 32-position representation statistic for text, not semantics.

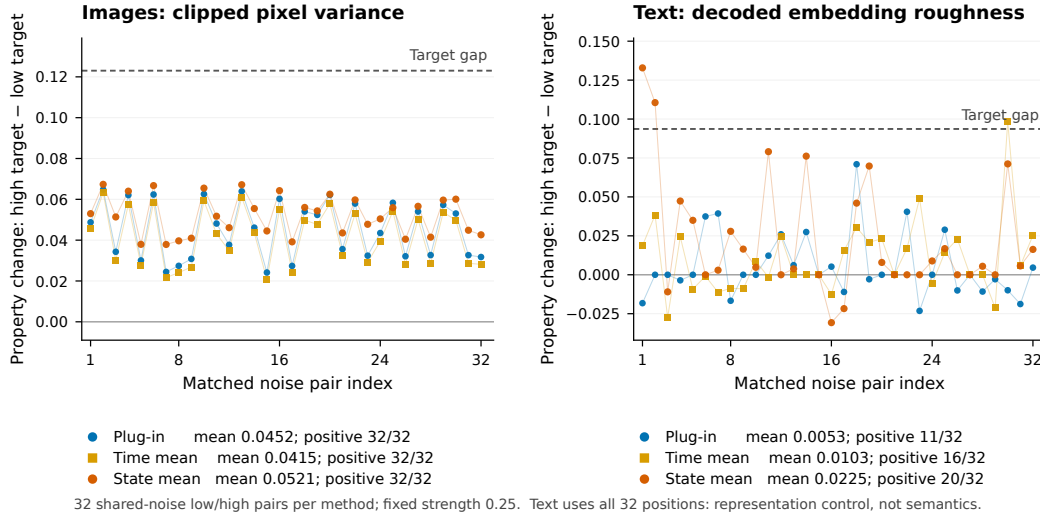


Figure 3: Within-noise high-minus-low target response. Each point is one counterfactual pair. The desired property difference is a reference, not an expectation that a finite-noise conditional sampler must match it exactly.

D Fixed-Index Qualitative Comparison

Final image examples · fixed indices 0-7, noise seed 39001

Seven methods; every displayed sample is fixed before inspection.

	#0 · low airplane $y=0.126$	#1 · high airplane $y=0.249$	#2 · low automobile $y=0.126$	#3 · high automobile $y=0.249$	#4 · low bird $y=0.126$	#5 · high bird $y=0.249$	#6 · low cat $y=0.126$	#7 · high cat $y=0.249$
Base	 $f=0.068$	 $f=0.080$	 $f=0.209$	 $f=0.209$	 $f=0.379$	 $f=0.207$	 $f=0.062$	 $f=0.200$
Post-hoc	 $f=0.126$	 $f=0.216$	 $f=0.126$	 $f=0.244$	 $f=0.126$	 $f=0.245$	 $f=0.126$	 $f=0.244$
Plug-in $\lambda = 4$	 $f=0.151$	 $f=0.282$	 $f=0.162$	 $f=0.284$	 $f=0.146$	 $f=0.274$	 $f=0.138$	 $f=0.269$
Time mean $\lambda = 4$	 $f=0.109$	 $f=0.255$	 $f=0.123$	 $f=0.260$	 $f=0.106$	 $f=0.245$	 $f=0.092$	 $f=0.242$
State mean $\lambda = 4$	 $f=0.132$	 $f=0.245$	 $f=0.125$	 $f=0.249$	 $f=0.131$	 $f=0.251$	 $f=0.118$	 $f=0.247$
State + variance $\lambda = 4$	 $f=0.123$	 $f=0.239$	 $f=0.131$	 $f=0.248$	 $f=0.139$	 $f=0.250$	 $f=0.106$	 $f=0.245$
Direct predictor $\lambda = 4$	 $f=0.131$	 $f=0.245$	 $f=0.124$	 $f=0.247$	 $f=0.129$	 $f=0.250$	 $f=0.116$	 $f=0.247$

Images are clipped to $[-1, 1]$; f is displayed-image variance. Same column shares noise and class across methods. Low/high columns use different noises. These examples show neither a quality ranking nor a matched-target intervention.

Figure 4: Final seed 39001, indices 0–7, fixed before inspection. Columns share noise and class across methods; low/high columns themselves have different noises. Every pixel example and property value is retained, including the ambiguous bird and cat. Residual and direct heads look similar; this small panel does not establish a quality ranking.

E Text Failure and Visible-Prefix Evaluation

For final seed 39001, sample 0, base and post-hoc decoding are exactly identical. The base’s raw property is 1.7703 and postprocessing changes it to the target 1.9446, but the decoded 32-position value remains 1.8900. This invariance holds for all 1,200 final text pairs. Equal-norm code vectors make nearest-vector decisions invariant to positive rescaling, up to numerical ties; it is a structural counterexample to evaluating only continuous error.

The complete sample-0 visible string is:

Footed , :le shouldatt and Vin Dr with "its – Ch route havesived , aism the New let class Br
" hous

The residual-head output is:

Footed , :ire shouldatt and Vin Dr with "its – Ch route havesivedu aism the New let class
Br " hous

The displayed dash is typeset from the original Unicode dash; the accompanying JSON preserves exact strings. Samples 0–2 of all seven methods are supplied without selection. These are fragmented outputs, regardless of PPL or property error. The full-property head produces the same visible sample-0 string as the residual head, reinforcing the limited evidence for residual structure.

A separate visible-prefix diagnostic truncates before the first EOS, removes token IDs 0/1/2, then measures adjacent-code-vector distance only when at least two tokens remain. It reports valid fraction, visible length, sequence-weighted and pair-weighted statistics. It does not score target success against the original full-position target: that would compare different properties. Detailed per-sequence values are in `analysis/moments/visible_prefix/`. Generated visible-prefix changes should not be called semantic control.

F Conditional Paired Uncertainty

The intervals below use 2,000 bootstrap replicates of paired examples within each observed noise seed. They concern the fixed generators and selected protocol, not a population of training runs or a selection-adjusted discovery claim. Negative differences favor residual mean calibration; the positive text difference against the direct head is retained.

Modality	Comparator	Output MAE difference	Conditional 95% interval
Images	Plug-in	−.02310	[−.02366, −.02256]
Images	Time mean	−.00466	[−.00513, −.00415]
Images	Direct head	−.00024	[−.00032, −.00015]
Text	Plug-in	−.06259	[−.06908, −.05650]
Text	Time mean	−.04289	[−.04847, −.03731]
Text	Direct head	+.00761	[+.00579, +.00947]

Table 4: Paired output-property effects over three sampling seeds, 400 samples each. Raw-state effects and seed-specific values remain in the analysis tables.

G Reproduction, Provenance, and Artifact Map

The independent project contains its own runtime, pretrained model assets, processed datasets, calibration inputs, and evaluation code. `calibration/` holds the 28-coefficient heads, property-only feature arrays, fitting code, and train/validation details. `sampling.py` implements the guidance field and Heun integration; `moment_benchmark.py` generates and evaluates the seven-method experiment. `runtime/` contains local model, decoder, dataset, and metric implementations. No external project directory is needed to run the experiment.

`runs/` retains raw states, targets, evaluator features, metrics, and model/fit digests. Results were preserved through a directory and metadata-schema reorganization; the import manifest retains source-record digests, while the underlying generated arrays and measured scores are unchanged. Refactored code is validated separately rather than described as the source of a new experiment. The supplied model assets were pretrained before this guidance study and remain frozen; the model card gives architecture, data, and training details without claiming retraining.

The README provides CPU pilot, head-fitting, final-experiment, analysis, and paper-build commands. Analytic tests check the Gaussian-bridge guidance factor, quadratic property gradients, correction derivatives, mean/variance ablations, norm caps, finite zero-field behavior, partial-batch counts, paired uncertainty, and decoding. Statistical summaries are generated from saved per-sample arrays. The verification record also checks that model and head values, sample arrays, and numerical results were preserved in the standalone package.